

AA/EE/ME 548

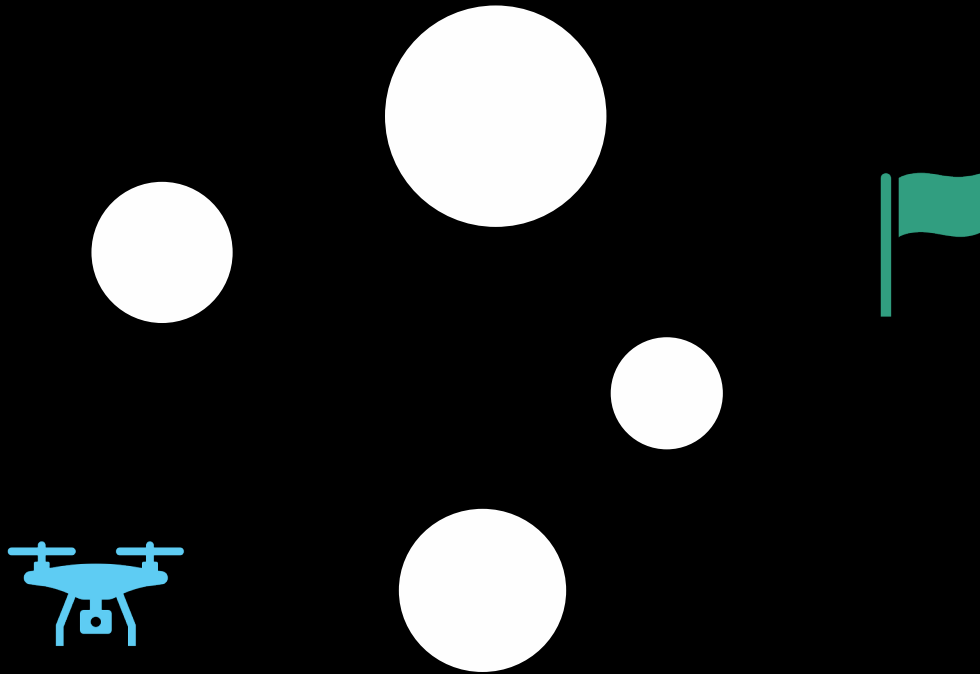
Linear Multivariable Control

Optimization- and learning-based, open-loop & closed-loop control

Spring 2026

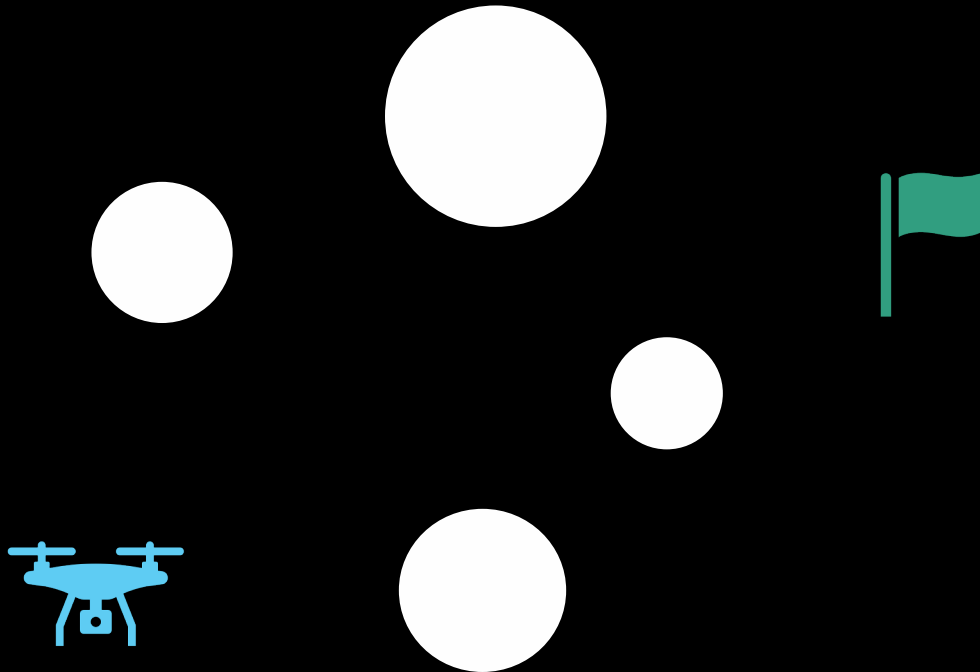


How to compute a control to perform desired task (e.g., navigate to goal)?

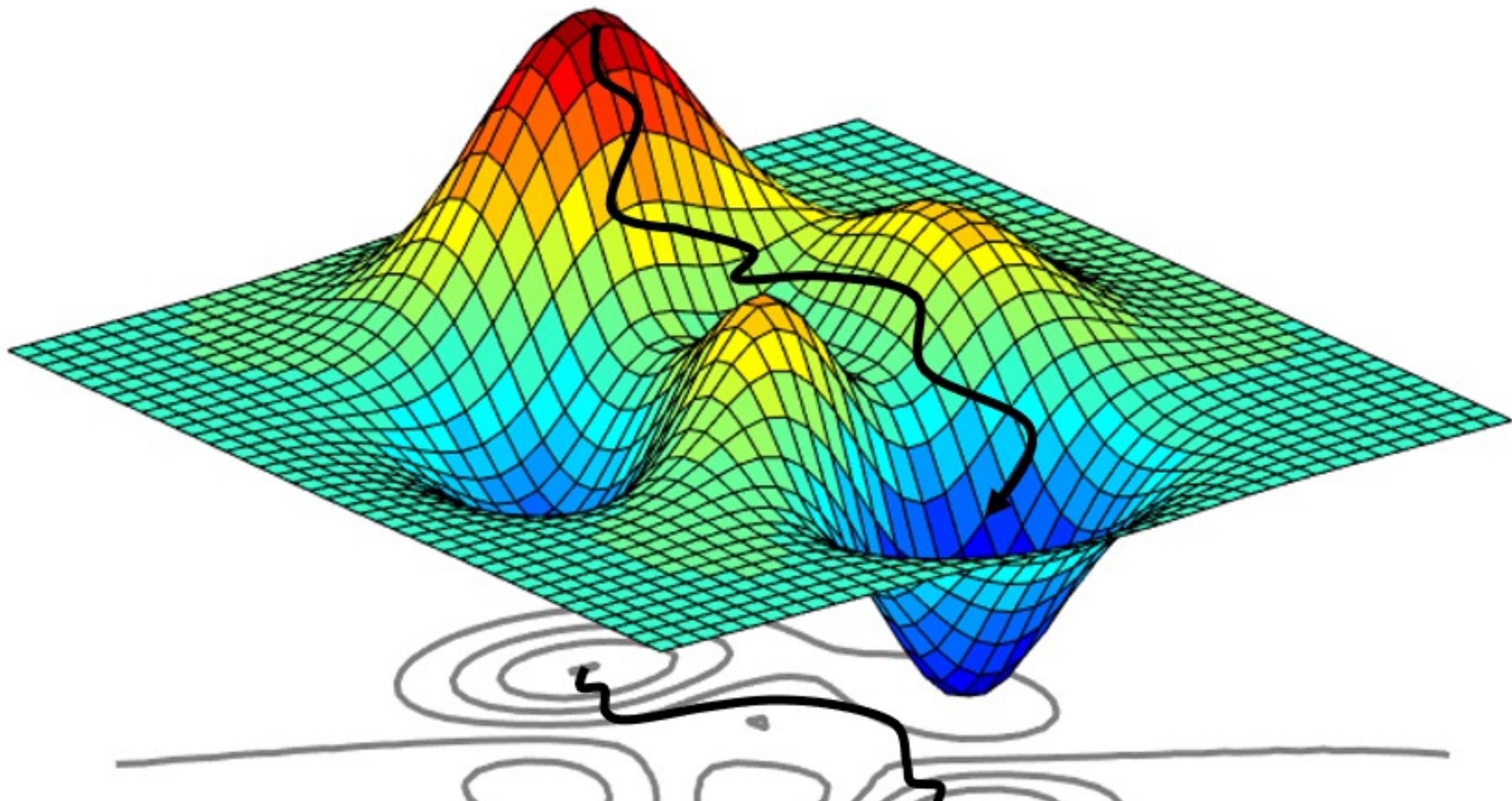


- Seek to know everything about the environment to make a *calculated* plan on what to do
- Use prior experience – apply *pattern matching* to obtain to decide on control
- Trial-and-error: try a bunch of things and see what works and doesn't work.

How to compute a control to perform desired task (e.g., navigate to goal)?



- Optimization-based control
Calculate plan on which to go
- Imitation learning
Learn from expert
- Reinforcement learning
Learn from trial and error



Optimization-based control

Assumed knowledge

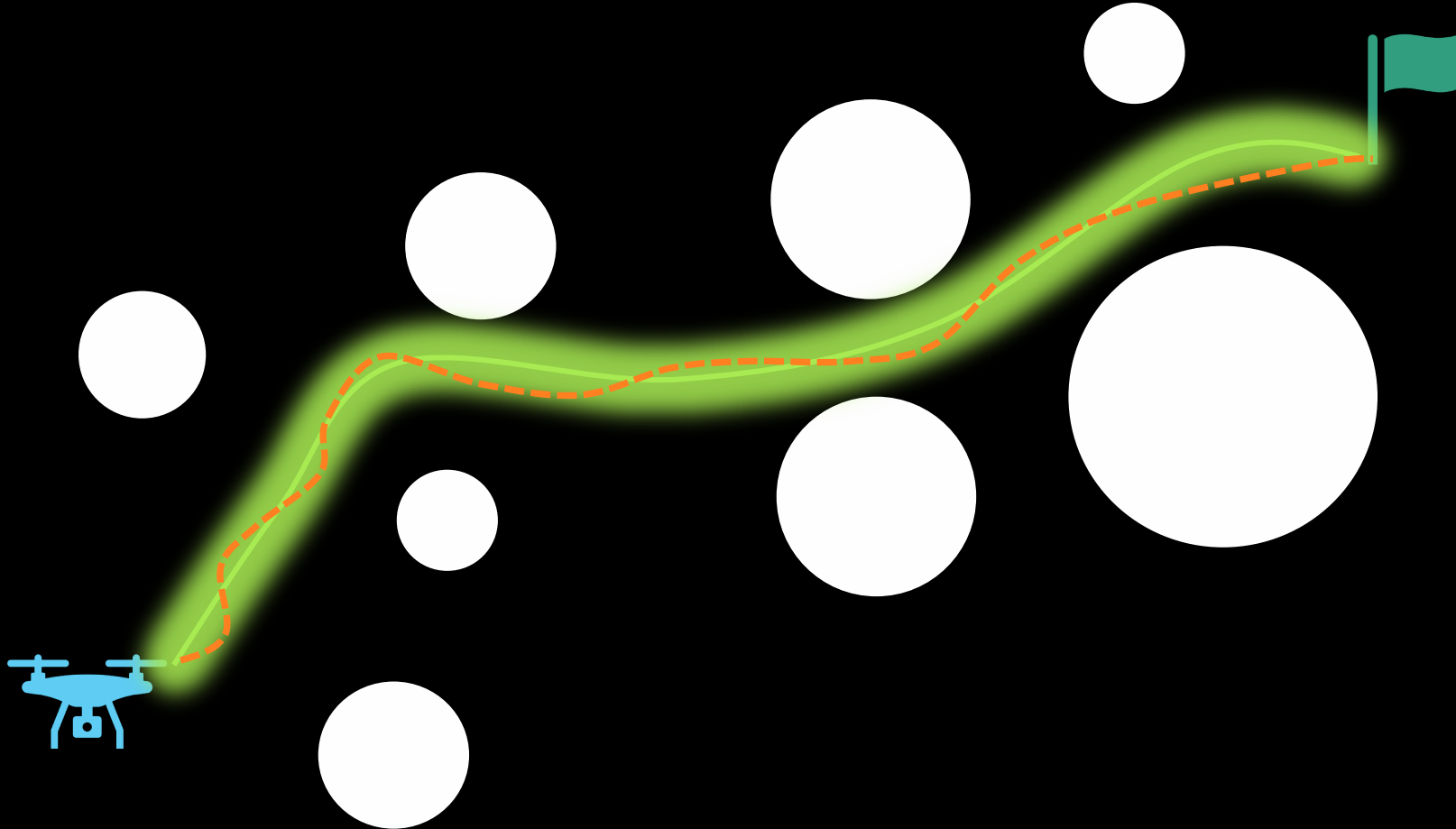
- System dynamics
 - $\dot{x} = f(x, u, d, t)$
- Environment information
 - Obstacle geometry and motion
- Task description
 - Objectives
 - Constraints

Optimization-based control: Pros and cons

Assumed knowledge

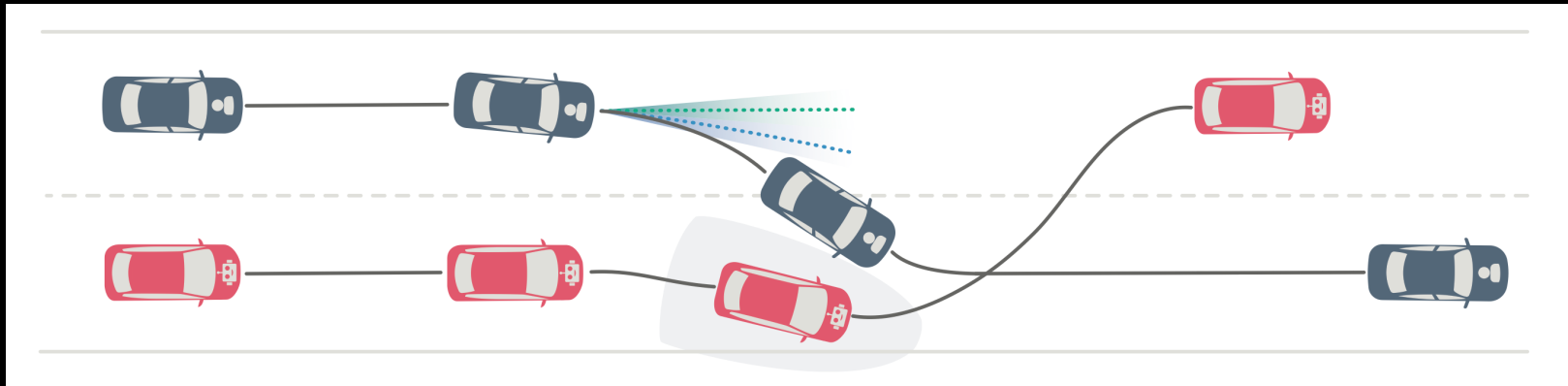
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Open-loop for high-level plan, closed-loop for tracking



Human-robot interactions: Incorporating human prediction within robot planning

Robots interacting with humans is challenging due to the uncertainty in human behaviors.





Imitation learning: Behavior cloning

Assumed knowledge

- Demonstrations of system performing the task

Idea

- Learn a policy that would reproduce the demonstration via *supervised learning*

Imitation learning: Generative modeling

Assumed knowledge

- Demonstrations of system performing the task

Idea

- Learn a policy that would reproduce the demonstrations by *matching the data distribution*

Imitation learning: Inverse optimal control

Assumed knowledge

- Demonstrations of system performing the task

Idea

- *Learn an objective function* that would reproduce the demonstrations

Imitation learning: Pros and cons

Assumed knowledge

- Demonstrations of system performing the task

Idea

- Learn the policy that would reproduce the demonstrations

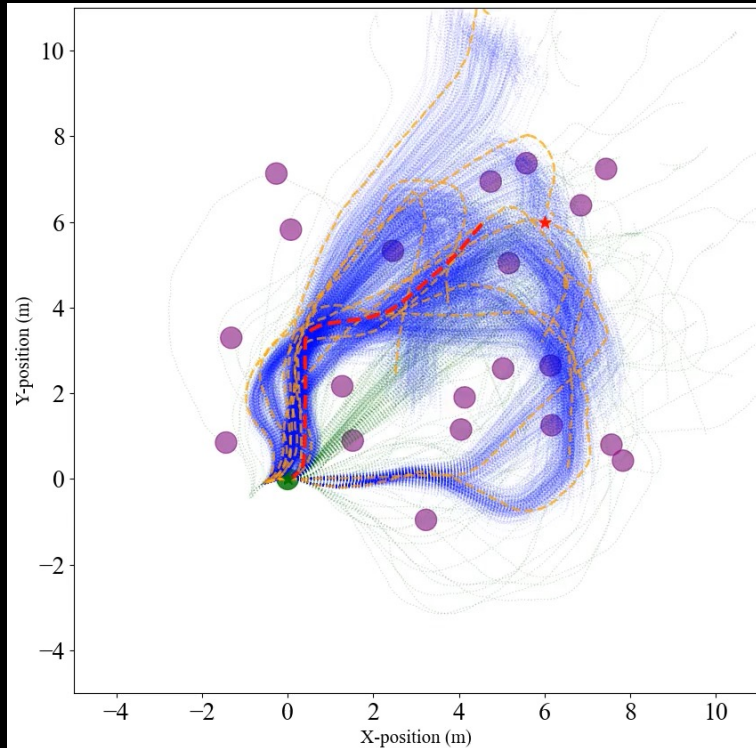


autonomous, 4x, 00:00

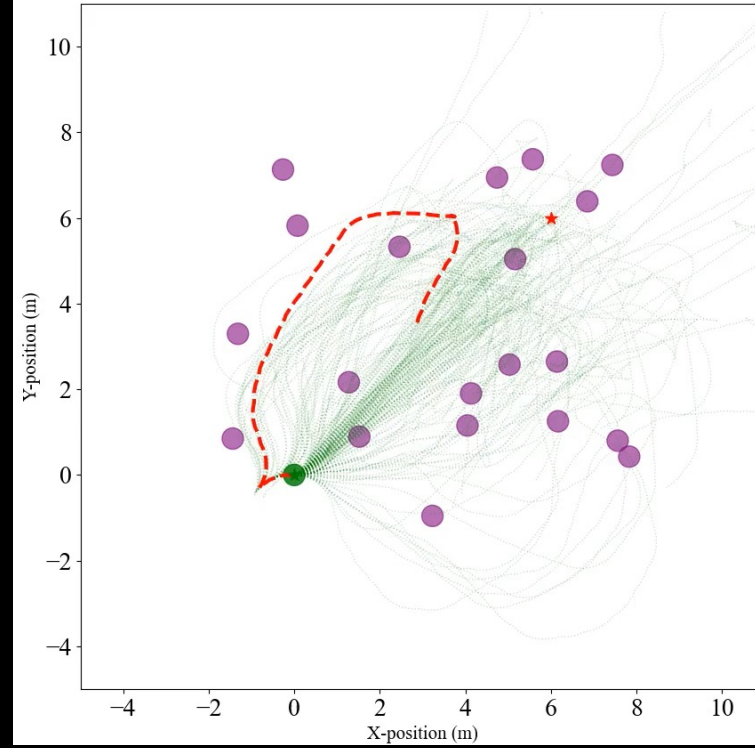
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Generation-refinement: Combining generative modeling with MPC

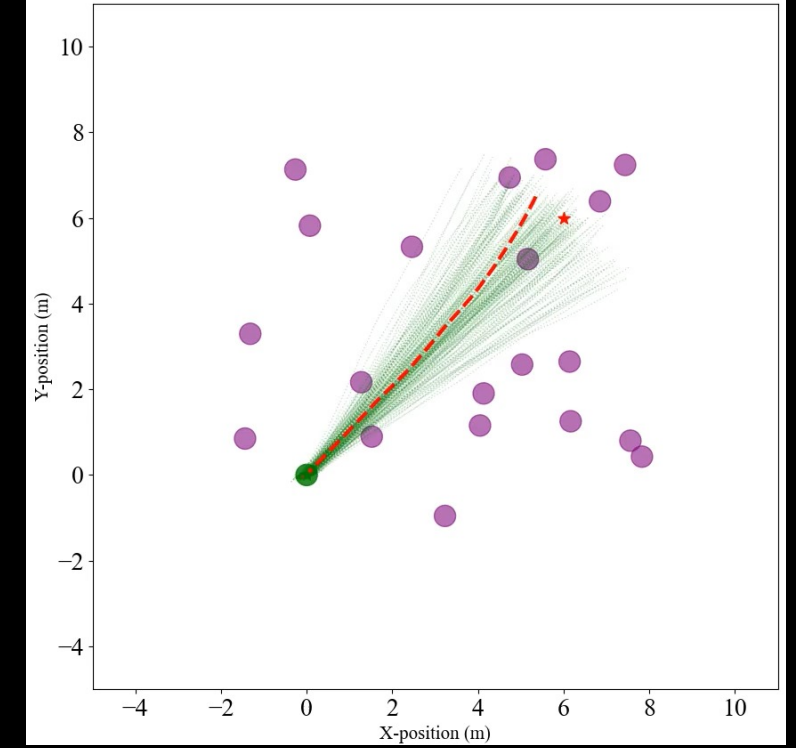
Generative model + optimization



Generative model only



Optimization only





Reinforcement learning

Assumed knowledge

- Interactions with the world
(observation, transition,
action)
- Reward signal

Reinforcement learning: Pros and cons

Assumed knowledge

- Interactions with the world
(observation, transition,
action)
- Reward signal

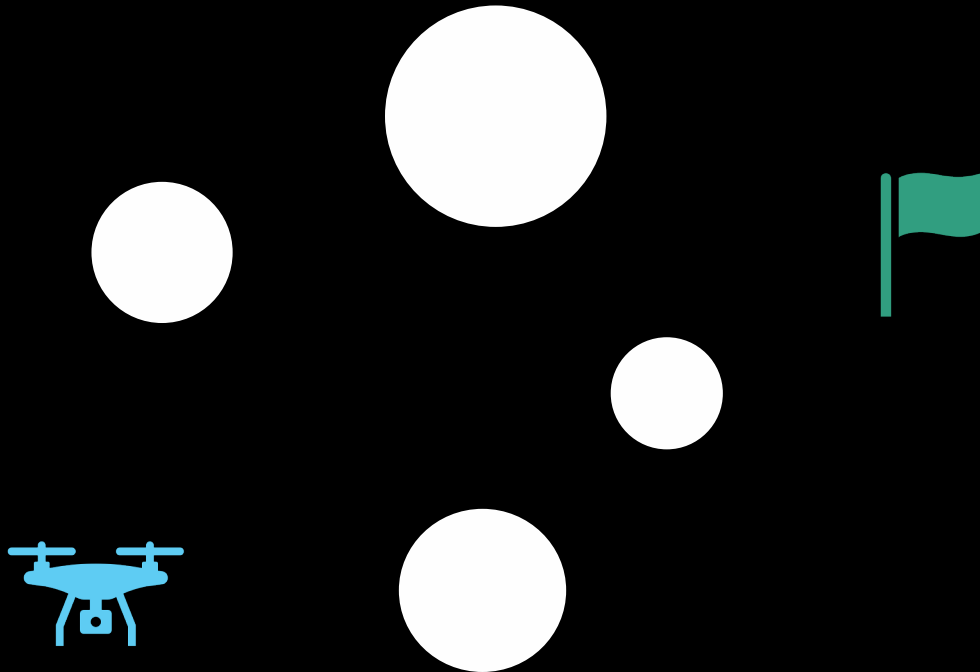


The new Atlas robot from Boston Dynamics

<https://www.youtube.com/shorts/DGlmZMTfFp8>



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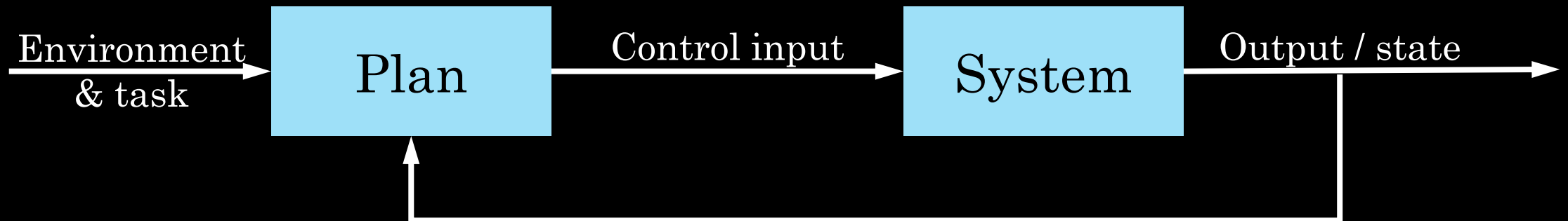
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Closed-loop vs open-loop control

Open-loop control: No feedback loop



Closed-loop control: No feedback loop



Open-loop control

Plan offline then execute with no updates online

E.g.,

Benefits

Limitations

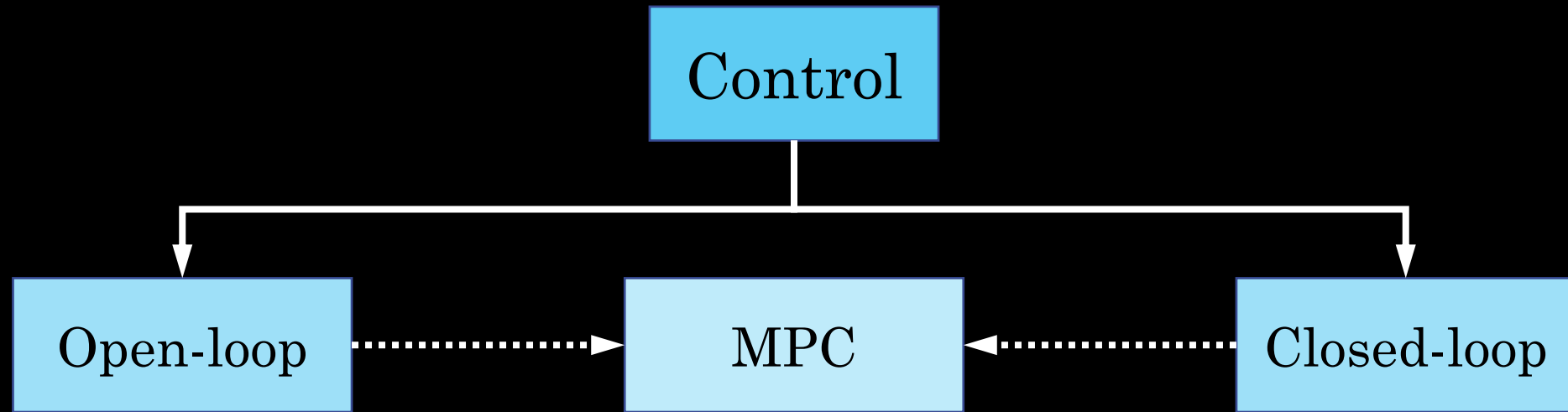
Closed-loop control

Execute controller $\pi(\cdot)$ online based on feedback on current output/state. Ideally, has insight into consequences of current actions

E.g.,

Benefits

Limitations



Model predictive control (MPC)

Closing the loop on open-loop control

Perform open-loop repeatedly online using updated output/state and environment information.

Model predictive **control**

